

Chapter 14

Recognition, Comprehension, and the Nonlinear Control Interpretation of Noticing

14.1 Introduction

In nonlinear control, the distinction between recognition and comprehension maps directly onto the difference between:

- Output-triggered anomaly detection, and
- State-space comprehension via nonlinear observers and invariant-set reasoning.

This chapter develops a nonlinear-systems interpretation of a powerful principle:

> If a controller or operator “notices” a problem, the nonlinear observer was not estimating the relevant state or manifold. The noticing event is evidence of incomplete system comprehension.

In nonlinear systems, noticing corresponds to a large deviation from the expected trajectory on the system's attractor or invariant manifold.

14.2 Recognition in Nonlinear Systems: Output-Level Surprise

Recognition corresponds to output-level detection in nonlinear systems—reacting only when the measured output $y(t)$ deviates significantly from the expected output trajectory.

14.2.1 Recognition as Output-Triggered Detection

For a nonlinear system:

$$\begin{aligned} & \dot{x} = f(x, u), \quad y = h(x) \end{aligned}$$

recognition occurs when:

$$|y(t) - \hat{y}(t)| = |h(x(t)) - h(\hat{x}(t))| > \epsilon$$

This is equivalent to:

- A threshold crossing
- A deviation from the nominal output manifold

- A mismatch large enough to be perceptible

Recognition is reactive and output-driven.

14.2.2 Recognition and Nonlinear Surprise

In nonlinear dynamics, noticing corresponds to:

- Leaving a stable manifold
- Entering a region with different local linearization
- Crossing a bifurcation threshold
- Encountering unmodeled nonlinearities

By the time the operator notices, the system has already moved into a new region of state space.

14.3 Comprehension in Nonlinear Systems: State-Space Understanding

Comprehension corresponds to accurate state estimation, manifold tracking, and predictive modeling of nonlinear dynamics.

14.3.1 Comprehension as Nonlinear State Estimation

A controller comprehends a nonlinear system when it maintains an internal estimate:

$$\hat{x}(t)$$

such that:

$$\hat{x}(t) \approx x(t)$$

and the estimation error:

$$e(t) = x(t) - \hat{x}(t)$$

remains small even under nonlinear transformations.

14.3.2 Comprehension as Manifold Tracking

For nonlinear systems, comprehension means:

- Tracking the system's invariant manifold
- Predicting motion along attractors
- Anticipating transitions between basins of attraction
- Detecting drift before it becomes output-visible

A system is comprehended when the observer reconstructs the true state trajectory before deviations propagate to the output.

14.4 Noticing as Nonlinear Estimation Error

14.4.1 The Nonlinear Control Meaning of Noticing

When an operator notices a problem, it means:

$$\|e(t)\| = \|x(t) - \hat{x}(t)\| \text{ is large}$$

The nonlinear observer failed to track the true state.

Thus:

> Noticing is a symptom of nonlinear observer failure.

14.4.2 Noticing as a Failure of Nonlinear Observability

If a problem becomes noticeable, then:

- The relevant state was not observable
- Or the observability rank condition was not satisfied
- Or the observer was not designed for the system's nonlinearities
- Or the model lacked the correct Lie derivatives

In control-theoretic terms:

$$\text{rank} \left(\frac{\partial}{\partial x} \right)$$

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\begin{bmatrix}
h(x) \\
L_f h(x) \\
L_{f^2} h(x) \\
\vdots \\
\end{bmatrix}
\right) < n
\}

```

14.4.3 Noticing as a Manifold-Departure Event

Noticing corresponds to:

- Leaving the nominal invariant set
- Entering a region with different local dynamics
- Exiting the observer's basin of attraction

A well-designed nonlinear observer prevents such departures.

14.5 Formalizing the Principle in Nonlinear Control

We can express your insight as a nonlinear-systems invariant:

> If an anomaly is noticed at time t , then the observer's estimation error was already non-zero at time $t^{\{-\}}$.

Formally:

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\{
\mathsf{Notice}(t) \rightarrow P(|e(t)| > \epsilon)
\}

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Where:

- $\mathsf{Notice}(t)$ = anomaly becomes salient
- $e(t)$ = estimation error
- P = previous time step

This is equivalent to saying:

> Noticing implies the nonlinear observer was not tracking the true state or manifold.

14.6 Implications for Nonlinear Control Design

14.6.1 Designing for Nonlinear Estimation, Not Recognition

High-performance nonlinear control systems must:

- Use nonlinear observers (e.g., EKF, UKF, high-gain observers)
- Model nonlinearities explicitly
- Track invariant manifolds
- Predict deviations before they manifest

Recognition-only systems rely on:

- Threshold alarms
- Output deviations
- Human noticing

These are insufficient for nonlinear stability.

14.6.2 Observer Design and Noticing

A well-designed nonlinear observer:

- Minimizes estimation error
- Detects drift early
- Flags anomalies before they reach the output
- Prevents operator surprise

A poorly designed observer forces the operator into a recognition-only mode.

14.6.3 Nonlinear Architecture Implications

Noticing indicates:

- Missing nonlinear terms in the model
- Incorrect Lie derivatives
- Unmodeled couplings
- Poor observability
- Incorrect manifold assumptions

In nonlinear control terms:

$$\mathcal{O} \neq \mathcal{S}$$

14.7 Applications Across Nonlinear Control Domains

14.7.1 Nonlinear Flight Dynamics

In aircraft:

- Noticing a stall is too late
- Comprehension requires nonlinear envelope protection
- Predictive observers prevent entry into unsafe regions

14.7.2 Chemical and Biological Nonlinear Systems

In process control:

- Noticing a runaway reaction means the observer failed
- Comprehension requires modeling nonlinear kinetics
- Predictive control prevents excursions

14.7.3 Robotics and Autonomous Nonlinear Systems

In robotics:

- Noticing instability means the controller was blind to nonlinear couplings
- Comprehension requires full-state nonlinear estimation
- Predictive control ensures stability margins

14.7.4 Human-in-the-Loop Nonlinear Control

When humans are part of the loop:

- Noticing indicates cognitive overload
- Comprehension requires shared nonlinear state estimation
- Predictive interfaces reduce operator surprise

14.8 Summary

This chapter reframes noticing as a nonlinear control signal:

- Recognition corresponds to output-level detection
- Comprehension corresponds to nonlinear state estimation
- Noticing indicates large estimation error
- Prevention requires nonlinear observers, not reactive alarms

The central nonlinear-systems principle:

> If a problem becomes noticeable, the nonlinear observer was not tracking the true state or manifold.

A fully understood nonlinear system does not surprise its controller.